

HUI ZHU

Ph.D. Student in Computer Science | Medical Image Analysis & Computational Biomedical Research
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RESEARCH PROFILE

Computer science Ph.D. student with research experience in medical image segmentation, multimodal learning, metastatic cancer imaging, and quantitative biomedical data analysis. Co-author of peer-reviewed work on unpaired multimodal medical image segmentation and longitudinal mechanical imaging of metastatic tumor spheroids. Experienced with cell-level analysis of microscopy images and with segmentation of structures in volumetric clinical imaging. Interested in applying AI and computational imaging methods to cutaneous disease, skin cancer, cellular pathology, and translational dermatology research.

RESEARCH INTERESTS

Medical image analysis | Image segmentation | Computational pathology | Multimodal learning | Cancer imaging | Microscopy image analysis | Biomedical signal processing | Trustworthy AI

EDUCATION

Ph.D. in Computer Science — Expected July 2028
Wayne State University, Detroit, Michigan

B.Eng. in Software Engineering — July 2023
Dalian University of Technology, Dalian, China

RESEARCH EXPERIENCE

Graduate Research Assistant, Trustworthy AI Lab — Aug. 2023 – Present
Department of Computer Science, Wayne State University | Advisor: Prof. Dongxiao Zhu

- Develop and evaluate deep learning methods for image segmentation, with particular experience in multimodal medical imaging and vision-language modeling.
- Co-authored MulModSeg, a method for unpaired multimodal medical image segmentation using modality-conditioned text embeddings and alternating training (WACV 2025).
- Contributed to research involving metastatic tumor spheroids and Brillouin microscopy, connecting computational analysis with cancer-cell mechanics and longitudinal imaging.
- Work in interdisciplinary teams spanning computer science, medical imaging, engineering, and biomedical research.

Undergraduate Researcher, Wang Laboratory — Sept. 2022 – July 2023
School of Software Engineering, Dalian University of Technology

- Processed experimental signal data and applied deep learning algorithms for feature extraction.
- Performed algorithm replication, parameter tuning, experimental reruns, and result analysis.

BIOMEDICAL IMAGING RESEARCH

Multimodal Segmentation of Clinical Imaging Across Unpaired Modalities

MulModSeg | IEEE/CVF WACV 2025, pp. 3581–3591

- Trained one segmentation model on **unpaired** CT and MR data, removing the usual requirement for spatially aligned scans from the same patient.
- Added modality awareness through modality-conditioned text embeddings from a frozen text encoder, plus alternating training to balance learning across modalities — no architectural change, no added inference cost.
- Evaluated on abdominal multi-organ and cardiac substructure segmentation with U-Net and Swin-UNETR backbones; consistent gains over prior methods on both modalities.
- *Relevance*: the same unpaired-multimodal structure arises when integrating tissue imaging with molecular assays collected from different patient cohorts.

Longitudinal Imaging and Classification of Metastatic Cancer Cells

Brillouin Microscopy of 3D Tumor Spheroids | Collaboration with the Zhang Lab (Biomedical Engineering)

- Analyzed label-free Brillouin microscopy images of cancerous and healthy cells in 3D culture — an optical modality reporting the mechanical properties of living cells without staining or fixation.
- Tracked cells across three sequential time stages to characterize how morphology and image-derived mechanical signatures evolved.
- Trained and evaluated a ResNet classifier separating cancerous from healthy cells directly from the microscopy images, against experimentally established ground truth.
- *Relevance:* cell-level classification from primary microscopy of living tissue models — the closest analogue in my work to tissue image analysis in pathology.

SELECTED PUBLICATIONS

1. **MulModSeg: Enhancing Unpaired Multi-Modal Medical Image Segmentation with Modality-Conditioned Text Embedding and Alternating Training.** WACV 2025. Chengyin Li, [Hui Zhu](#), Rafi Ibn Sultan, Hassan Bagher Ebadian, Prashant Khanduri, Chetty Indrin, Kundan Thind, Dongxiao Zhu.
2. **Longitudinal Mechanical Imaging of Metastatic Tumor Spheroids Using Brillouin Microscopy.** Optical Elastography and Tissue Biomechanics XII. Karlin Hilai, Daniil Grubich, Marcus Akrawi, [Hui Zhu](#), Razanne Zaghloul, Chenjun Shi, Man Do, Dongxiao Zhu, Jitao Zhang.
3. **Mechanical Evolution of Metastatic Cancer Cells in Three-Dimensional Microenvironment.** K. Hilai, D. Grubich, M. Akrawi, [H. Zhu](#), R. Zaghloul, C. Shi, M. Do, D. Zhu, et al.
4. **GeoSAM: Fine-Tuning SAM with Sparse and Dense Visual Prompting for Automated Segmentation of Mobility Infrastructure.** R. I. Sultan, C. Li, [H. Zhu](#), P. Khanduri, M. Brocanelli, D. Zhu.
5. **Automatic Calibration for Membership Inference Attack on Large Language Models.** ECAI. Saleh Zare Zade, Yao Qiang, Xiangyu Zhou, [Hui Zhu](#), Mohammad Amin Roshani, Prashant Khanduri, Dongxiao Zhu.

TEACHING & LEADERSHIP

Teaching Assistant / Project Instructor, CSC 4996 Capstone Projects — Jan. 2024 – Present

Wayne State University

- Guide student teams through software project design, implementation, milestones, and presentation.
- Supported projects including Backpain Image Analysis, Dealership Compliance Auditor, Voice-to-Song, Footprint, Voice-Controlled Autonomous Car, and SnapMechanic.

Algorithm Analysis Team Lead — May 2022

- Coordinated literature review, code reproduction, parameter studies, experiment monitoring, and project progress.

TECHNICAL SKILLS

Programming: Python, C/C++, Bash, SQL, LaTeX

Deep Learning Frameworks: PyTorch, MONAI, Hugging Face Transformers, scikit-learn, NumPy, SciPy, pandas

Architectures & Methods: U-Net and encoder-decoder segmentation, ResNet and CNN classifiers, Vision Transformers (Swin-UNETR), vision-language models, Segment Anything Model (SAM) fine-tuning, multimodal and text-conditioned learning, transfer learning

Medical & Biological Imaging: 2D/3D volumetric segmentation (CT, MR), microscopy image analysis, DICOM and NIfTI handling, preprocessing pipelines (resampling, cropping, padding, intensity normalization), data augmentation, OpenCV, scikit-image

Evaluation & Rigor: Dice, IoU and boundary-based segmentation metrics, ROC/AUC and classification metrics, cross-validation, ablation design, model calibration and uncertainty estimation, reproducible experiment tracking

Compute & Tooling: Linux, Git/GitHub, Docker, CUDA and multi-GPU training, HPC cluster job scheduling, Conda environments